

NFL Big Data Bowl

NYU Edition

The Challenge

Build a data tool or visualization that an NFL coach, scout, or broadcaster would use.

Using real NFL Next Gen Stats tracking data, discover an insight from the data and bring it to life through code. You have **3 hours**. Kiro (AI-powered IDE) is available to accelerate your exploration and development.

Your Task

Using the provided dataset (NFL skill-position player tracking data from the 2021 season), build something that surfaces a football insight about TEs, WRs, or RBs. You choose the format:

- **Visualization or Comparison Tool** - A chart, dashboard, animation, or interactive tool that reveals a pattern or lets users compare players/matchups/tendencies
- **Metric or Score** - A new statistic you've invented that quantifies something not currently measured (e.g., a "separation score," a "route efficiency index," a "yards after contact predictor")

Requirements

- **1. An original insight** - Your output must reveal something a football professional would care about, focused on skill positions (TE/WR/RB)
- **2. Working code** - Built in Python. Kiro is available as an AI-assisted development tool to accelerate your work.
- **3. A README** (3-5 sentences) - What you built, what insight it reveals, and who would use it (coach? scout? broadcaster?)

What You're Working With

The Data

- **Source:** 2021 NFL season, Weeks 1-8
- **Tracking data:** X/Y coordinates at 10 frames per second for all 22 players per play
- **PFF Scouting data:** Player role assignments, matchup pairs, and outcome labels
- **Focus:** Skill positions - TE, WR, RB routes, targets, and yards after catch

Provided

Files/folders	Description
`tracking/`	Per-game frame-by-frame tracking (for 122 games): player + ball x/y, speed, acceleration, orientation, direction, events.
`games.csv`	Game-level metadata for Weeks 1–8 (122 games) incl. season, week, teams
`pffScoutingData.csv`	PFF role/matchup annotations with outcome labels
`players.csv`	Player info (name, position)
`plays.csv`	Play-level metadata (down, distance, formation, coverage)

The Tools

- **Kiro** - AI-powered IDE. Use it to explore, generate code, debug errors, get analysis suggestions, refactor, and generate documentation. Access here: https://kiro.dev/students/#instructions
- **Python libraries:** pandas, numpy, matplotlib, seaborn, plotly, scikit-learn, scipy

Helpful Context

- `playDirection` is 'left' or 'right' (normalize coordinates so all plays go the same direction)
- Pass-block window = ball_snap to first of {pass_forward, qb_sack, qb_strip_sack, run}
- Tracking for deep play-level analysis + PFF data for cross-week context
- `passResult` values: C (complete), I (incomplete), S (sack), R (scramble), IN (interception)
- `pff_role` values include: Pass Route, Coverage, Pass Block, Pass Rush, Pass

How You'll Be Judged | Submission deadline: 2:00 PM (sharp)

What to Submit

A hosted solution via one URL with references to:

- Your code (notebook or .py files)
- Your output (visualization, tool screenshot, or metric demonstration)
- Your README (3-5 sentences) including findings, methods, assumptions, and limitations.

How Judging Works

Judges will review all submissions as follows:

1. **Triage:** Judges will split up and evaluate submissions based on: 1) Does it run? 2) Is there a clear insight? 3) Does the README make sense?
2. **Rubric Grading :** Top submissions from the Triage stage are scored 1-5 on Football Insight, Technical Execution, Communication, and Originality (below) Winners are selected based on total score.

Dimension (1-5)	What judges look for
Football Insight	Does your finding matter? Would a coach, scout, or broadcaster use it?
Technical Execution	Does the code work? Clean? Efficient? Good use of libraries?
Communication	Is the output clear? Does the README explain the value? Is the visualization readable?
Originality	Non-obvious approach? Creative framing? Something new (not already discovered as part of the 2023 Big Data Bowl)?

Tiebreaker order: Football Insight > Technical Execution > Originality > Revote

Awards

- **1st Place** - Overall highest score, wins NFL game tickets + AWS interview
- **2nd Place** - Second highest score, wins NFL game tickets

What to Submit

- Your code (notebook or .py files)
- Your output (visualization, tool screenshot, or metric demonstration)
- Your README (3-5 sentences)

How to submit

- https://pulse.amazon/survey/23X4OOUL

Submission deadline: 2:00 PM (sharp)

Tips

- **Don't boil the ocean.** Pick ONE player, ONE route concept, or ONE question and go deep.
- **Use Kiro.** Let it generate your exploratory analysis, data loading code, plotting boilerplate, and documentation. Spend YOUR time on the insight. https://kiro.dev/students/#instructions
- **Think about your audience.** A coach wants actionable recommendations. A broadcaster wants a visual story. A scout wants comparative data.
- **Show, don't tell.** A compelling visualization beats a wall of text.
- **The README/format matters.** Judges read it first. Make it clear what you built/why it's interesting.

Run of Show

Time	Activity
10:00 AM	Venue Opens; Participant Check-ins and Breakfast
10:30 AM	Opening Remarks and Competition Directions
11:00 AM	**Challenge Begins**
2:00 PM	**Challenge Ends: All Submissions Due**
2:00 PM	Lunch & Judging Begins
3:00 PM	Announcement of Winners
3:15 PM	Event Ends

Pre-Event Setup

- Bring your laptop, charger, and headphones

Rules

- Code of Conduct. All participants agree to keep this a safe, welcoming, and inclusive space for everyone.
- Eligibility. This event is open to registered participants only. You must have received an invitation and completed the RSVP process to participate. Organizers, volunteers, judges, sponsors, or anyone in a privileged position at the event may not participate as a competitor.
- Team Composition. Teams of up to 3 members are permitted. All team members must be present at the event and actively contribute to the project. Solo participation is also allowed.
- All work must be done on-site. All coding, analysis, and visualization work must be completed during the 3-hour competition window. You may not use pre-built code, notebooks, or project materials created before the event.
- Ideas are fair game. You may come in with an idea or hypothesis you'd like to explore - that's encouraged. However, no code, visualizations, or written analysis may be prepared in advance.
- Dataset. You must use the provided dataset. You may also incorporate external datasets, provided they are publicly available. The dataset and challenge prompt will be distributed at the start of the
- Submission Format. Each team must submit a public repository with:

    - Working code
    - A README (3-5 sentences) describing your insight/work
    - Your visualization or metric/score tool output
- Open-Source Libraries Allowed. Teams may use any publicly available libraries, frameworks, or packages. Pre-building a library specifically for this event and open-sourcing it is not permitted.
- Originality. Your submission must represent original work produced at this event. Submitting work from a prior project, class assignment, or previous competition is not allowed.
- Disqualification. Teams may be disqualified at the organizers' discretion for violating these rules, the Code of Conduct, or engaging in unsporting behavior.

Good luck. Find something the NFL hasn't seen yet!